Appl Phys A (2008) 93: 1011–1014
DOI 10.1007/s00339-008-4786-6

Applied Physics A
Materials Science & Processing

Laser-assisted local patterning of ZnO-based spots for mirror-less lasing

A.V. Kabashin · W. Marine · M. Meunier

Received: 12 October 2007 / Accepted: 9 April 2008 / Published online: 13 July 2008
© Springer-Verlag 2008

Abstract ZnO is known as one of the best materials for the implementation of the random lasing effect, associated with mirror-less laser emission in a simultaneously amplifying and highly scattering medium. Normally, the fabrication of this medium requires a rather complicated procedure of deposition and thermal treatment of ZnO-based films on some specific substrates, yielding wurtzite-orientation ZnO nanocrystals. We demonstrate a rapid synthesis of highly efficient ZnO-based random lasing spots on a piece of Zn by employing the phenomenon of laser-induced air breakdown. Being ignited near the surface of a Zn target, plasma of the air breakdown serves as a local reactor to locally transform its properties and thus form a film of well-packed 20–40 nm ZnO nanospheres. Exhibiting extremely high amplification and scattering, this medium is capable of generating the random lasing effect within the exciton-based photoluminescent band.

PACS 81.16.Mk · 81.05.-t

1 Introduction

Still looking rather futuristic, mirror-less lasers can become feasible due to recent observations of the random lasing effect in highly disordered media [1–10]. When the scattering mean free path is equal to or less than a wavelength, a photon may return to a scatterer from which it was scattered before, forming closed-loop paths. If the amplification along such a path exceeds the loss, laser oscillation can arise in the 'self-formed' loop, which serves as a mirror-less laser resonator [11]. Thus, instead of bouncing from one mirror to another, the light waves bounce from one particle to another many times before they leave the disordered material, as shown in Fig. 1, providing narrow laser emission lines and offering spectacular opportunities for the development of novel optoelectronics devices and components.

Combining a strong scattering and a high gain due to an extremely efficient direct band gap transition at 3.37 eV, nanostructured ZnO offers an ideal medium for the implementation of the random lasing effect [3–8]. However, since it requires the formation of nanoscale, well-separated nanocrystals [12], the fabrication of such a medium is not easy. Normally, this can be achieved only through a combination of deposition of a highly stoichiometric ZnO film from a ZnO target on some specific substrates (sapphire, silica, silicon) and its annealing to achieve a wurtzite structure of constituent crystals [4–8]. In this case, the random lasing effect is observed by the appearance of several narrow (less than 0.3 nm full width at half-maximum (FWHM)) laser lines in the exciton emission band (380–385 nm).

In this paper, we show that the ignition of laser-induced breakdown of air near a Zn target can lead to a synthesis of an extremely efficient amplifying and scattering medium under the irradiation spot, which can be used for mirror-less lasing. Although this paper does not consider a detailed analysis of properties of the produced structures (it will be published elsewhere), it gives an idea of the essence of the breakdown-based treatment and its difference compared to conventional laser ablation.

A.V. Kabashin (✉) · M. Meunier
Engineering Physics Department, Ecole Polytechnique de Montréal, Montréal, Canada, H3C 3A7
e-mail: andrei.kabashin@polymtl.ca

W. Marine
Département de Physique, Faculté des Sciences de Luminy, CRMCN UPR CNRS 7251, 13288 Marseille, France

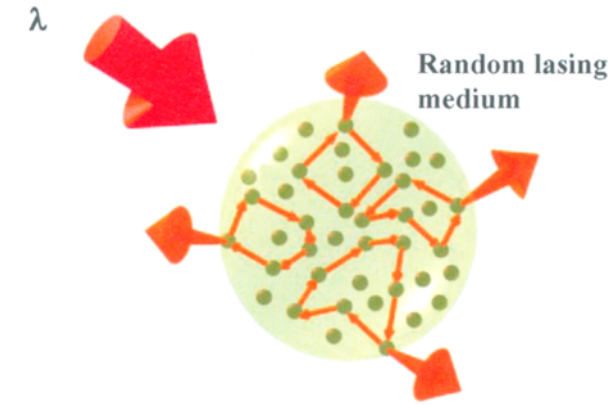

Fig. 1 Mechanism of random lasing in highly disordered media; photons are amplified in closed-loop paths forming mirror-less laser resonators

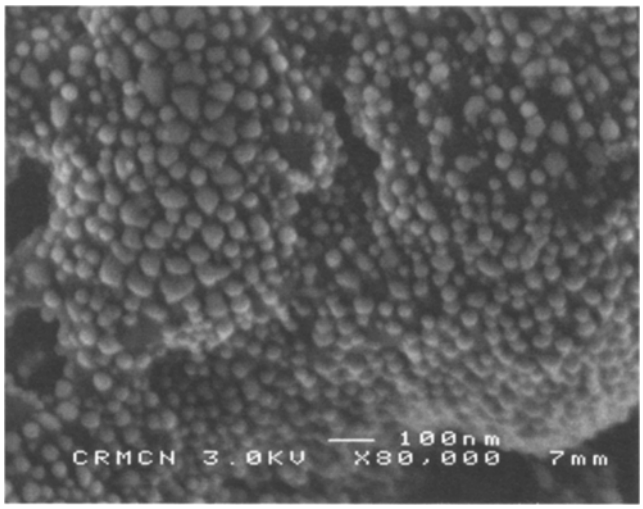

Fig. 2 Typical scanning electron microscopy image of Zn treated by laser-induced air breakdown processing

2 Results and discussion

It is known that the breakdown of air can be initiated by a simple focusing of an infrared (IR) laser beam in air [13]. The installation of a target near the lens focus leads to a two orders of magnitude decrease of the breakdown threshold [14, 15]. It is important that unlike conventional laser ablation, the breakdown is accompanied by a strong absorption of the incoming radiation by the breakdown plasma itself, leading to extremely high temperatures and ultra-intense electric currents within the plasma region [13, 15–18]. The target serves to generate initial electrons and the formed plasma develops in the cold gas toward the focusing lens and absorbs the main IR radiation power at its shock wave forefront through the inverse Bremsstrahlung mechanism [13]. As a result, the plasma becomes heated to temperatures of about 10^4 K, while the radiation power on the target surface decreases dramatically and the radiation-related ablation of material drops. We use plasma of laser-induced air breakdown to transform properties of a Zn target.

In the experiments, the breakdown was created in ambient air near the surface of a Zn foil by focused radiation from a pulsed TEA CO_2 laser (wavelength 10.6 μm, pulse energy 1 J, typical pulse shape with a leading 200 ns FWHM spike and 1 μs host, repetition rate 3 Hz). Standard targets of metallic Zn (purity 99.995%) with dimensions of about 1×1 cm^2 were used as targets and the focal size on the target surface was 1.5×1 mm^2. Experiments were carried out in atmospheric air (1 atm, 20°C, 40% humidity). Photoluminescence (PL) spectra were recorded on a double spectrometer (U100, Instruments SA) using 325-nm radiation of a $HeCd^+$ laser as a source and a GaAs photomultiplier as the detector. Scanning electron microscopy (SEM) was used to examine structural properties of the films. In addition, X-ray photoemission spectroscopy (XPS) was used to determine surface layer composition.

The breakdown manifested itself as a bright flash, accompanied by sound effects, moving toward the focusing lens and reaching distances of about 1 cm from the target. The breakdown treatment by several pulses led to the formation of a grey area on the target surface. Despite a rather large distance of plasma propagation, only the area under the irradiation spot was affected by the breakdown treatment, suggesting that the ablated material always returns due to hydrodynamic processes in the breakdown plasma. Nevertheless, no crater was present on the target surface, as takes place in experiments on ultraviolet or femtosecond laser ablation. The absence of the crater illustrates a radical difference of the mechanism of the breakdown treatment compared to conventional laser ablation, which is characterized by plasma-based rather than target material-based absorption of radiation energy. Figure 2 shows typical scanning electron microscopy images of the treated area. One can see that the produced layers on Zn were composed of 20–40 nm spheres, forming a well-packed 'photon-crystal-like structure'. The size of the spheres slightly increases to 40–50 nm when the intensity of the breakdown was increased through the increase of CO_2 laser energy. In addition, the spheres formed microscale 'cauliflower'-like agglomerations at relatively high laser fluences. The structural analysis, performed by X-ray photoelectron spectroscopy (XPS), revealed the presence of almost stoichiometric ZnO in the nanoparticle composition. The presence of a strong (002) peak at $2\theta = 34.4°$deg. in the X-ray diffraction spectra (inset to Fig. 3) suggested the presence of sub-micron ZnO crystals with wurtzite structure with their c axis normal to the substrate plane. Measuring the broadness of the XRD peak, we can estimate, with a fair accuracy, the minimal size of spherical nanoparticles in the treated area by the Debye–Scherrer formula [19]. Results of such estimation, presented

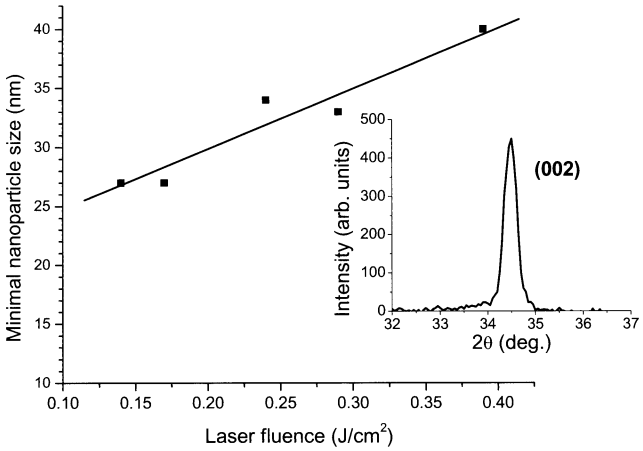

Fig. 3 Minimal size of nanoparticles in the treated area as a function of the laser fluence used for the breakdown initiation. *Inset*: typical highly resolved X-ray diffraction pattern near (0-0-2) line from a Zn target

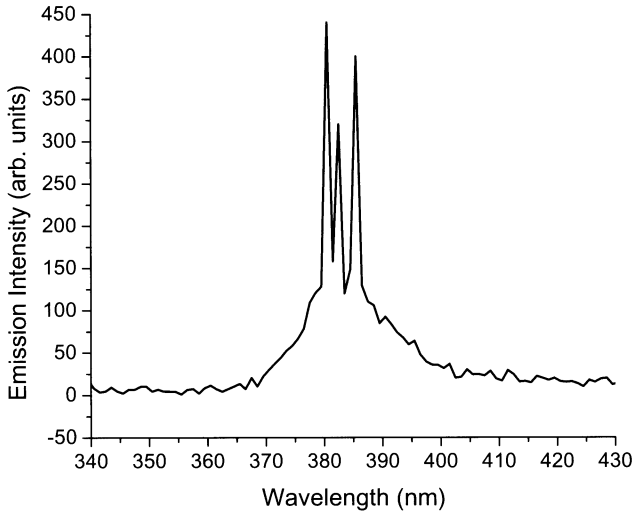

Fig. 4 High resolution photoluminescent spectrum band showing several narrow laser lines within the exciton emission band

in Fig. 3, suggest that the nanoparticle size slightly increases under the increase of CO_2 laser energy (from 27 to 40 nm), which was consistent with the above-stated visual observations of the nanosphere size by SEM.

The treated area exhibited striking photoluminescence properties. First, the exciton band at 380–385 nm was very strong, while defect-related bands were extremely weak and completely disappeared under a relatively high fluence of breakdown-pumping radiation (details of behavior of different photoluminescence bands will be published elsewhere). However, the most interesting effect was related to the generation of several narrow laser lines (the width of these lines was less than 0.5 nm) within the exciton emission band, as shown in Fig. 4, and the number of the lines increased as the intensity of photoluminescence-pumping radiation increased. The generation of such lines is normally attributed to the random lasing effect described above [3–7]. Thus, we were able to synthesize a random lasing medium in one step, using a piece of Zn as a starting material.

Despite the relative simplicity of the breakdown phenomenon, the mechanism of formation of the photon-crystal-like nanostructures exhibiting the random lasing effect is not yet clear. We believe that, once ablated, the material from a target finds itself in a hot plasma 'reactor', drastically transforming its properties. The material is then re-deposited within the irradiation spot forming densely packed nanospherical structures, as shown in Fig. 2. However, other mechanisms cannot be ruled out. In particular, we cannot exclude another mechanism implying surface erosion without any flash evaporation of material and its re-deposition. Clarification of conditions and parameters of material transformation is a subject of our on-going project. These results will be published elsewhere. It should be noted that the synthesized random lasing medium structure is impossible to produce by conventional laser ablation (at least, without any additional step). The main problem with laser ablation is related to a rapid release and cooling of ablated species under their interaction with the environment, which prevents the formation of the required nanostructure and stoichiometry. In contrast, laser-induced air breakdown processing makes possible the nanoparticle treatment in a 'plasma reactor', combining extremely high temperatures with a chemically active environment. This feature makes the breakdown processing unique and different from any other laser-assisted technique.

3 Conclusions

In summary, we were able to produce ZnO-based random lasing spots on a Zn target in air using a technique of laser-induced air breakdown processing. The produced layers exhibited defect-free emission of an exciton-based band with narrow lines associated with the random lasing effect. Due to its extreme simplicity and cost efficiency, the proposed method can be of importance for local patterning of random lasing components for optoelectronics applications.

The authors acknowledge the financial contributions from the Natural Science and Engineering Research Council of Canada and from the Canada Research Chairs.

References

1. N.M. Lawandy, R.M. Balachandran, A.S.L. Gomes, E. Sauvain, Nature (Lond.) **368**, 436 (1994)
2. D.S. Wiersma, M.P. Van Albada, A. Lagendijk, Phys. Rev. Lett. **75**, 1739 (1995)

3. H. Cao, Y.G. Zhao, H.C. Ong, S.T. Ho, J.Y. Dai, J.Y. Wu, R.P.H. Chang, Appl. Phys. Lett. **73**, 3656 (1998)
4. Z.K. Tang, G.K.L. Wong, P. Yu, M. Kawasaki, A. Ohtomo, H. Koinuma, Y. Segawa, Appl. Phys. Lett. **72**, 3270 (1998)
5. H.C. Ong, J.Y. Dai, A.S.K. Li, G.T. Du, R.P.H. Chang, S.T. Ho, J. Appl. Phys. **90**, 1663 (2001)
6. H.D. Li, S.F. Yu, S.P. Lau, E.S.P. Leong, Appl. Phys. Lett. **89**, 021110 (2006)
7. S.F. Yu, C. Yuen, S.P. Lau, H.W. Lee, Appl. Phys. Lett. **84**, 3244 (2004)
8. E.V. Chelnokov, N. Bityurin, I. Ozerov, W. Marine, Appl. Phys. Lett. **89**, 171119 (2006)
9. S.V. Frolov, W. Gellerman, M. Ozaki, K. Yoshino, Z.V. Vardeny, Phys. Rev. Lett. **78**, 729 (1997)
10. G.D. Dice, S. Mujumdamar, A.Y. Elezzabi, Appl. Phys. Lett. **86**, 131105 (2005)
11. D.S. Wiersma, A. Lagendijk, Phys. Rev. E **54**, 4256 (1996)
12. U. Ozgur, Y.I. Alilov, C. Liu, A. Teke, M.A. Reshnikov, S. Dogan, V. Avrutin, S.-J. Cho, H. Morkoc, J. Appl. Phys. **98**, 041301 (2005)
13. Yu.P. Raizer, *Laser-Induced Discharge Phenomena* (Consultants Bureau, New York, 1977)
14. F.V. Bunkin, V.I. Konov, A.M. Prokhorov, V.B. Fedorov, JETP Lett. **9**, 371 (1969)
15. A.M. Prokhorov, V.I. Konov, I. Ursu, I.N. Mihailescu, *Laser Heating of Metals* (IOP, UK, 1990)
16. A.V. Kabashin, M. Meunier, Appl. Surf. Sci. **186**, 578 (2002)
17. A.V. Kabashin, M. Meunier, Appl. Phys. Lett. **82**, 1619 (2003)
18. A.V. Kabashin, M. Meunier, Mater. Sci. Eng. B **101**, 60 (2003)
19. B.D. Cullity, *Elements of X-ray Diffraction* (Addison–Wesley, Reading, 1978)